First interrogation of Algebra I
November 2024 (60 mn)

Exercice 1 .(04 Pts)

1. Show by induction : $\forall n \in \mathbb{N} : \sum_{k=0}^{n} 2^k = 2^{n+1} - 1$;**(02 Pts)**
2. Show by contrapositive : n prime $\Rightarrow n = 2$ or n odd.**(02 Pts)**

Exercice 2 .(04 Pts)

Let $\Re$ the relation in $\mathbb{R}$ defined by :

$$a \Re b \Leftrightarrow a^3 - b^3 = a - b$$

1. Show that $\Re$ is an equivalence relation.**(02 Pts)**
2. Discuss according to m the number of elements in the class of m.**(02 Pts)**

Exercice 3 .(12 Pts)

Let $S \neq \emptyset$, $A, B \in \mathcal{P}(S)$. We define a function f as :

$$\begin{array}{rccc} f & : & \mathcal{P}(S) & \to & \mathcal{P}(A) \times \mathcal{P}(B) \\ & & X & \mapsto & (X \cap A; X \cap B) \end{array}$$

1. Let $A = \{e, \{f, g\}\}, B = \{d\}$. Calculate the power set of A, and the Cartesian product of $\mathcal{P}(A)$ and $\mathcal{P}(B)$. **(03 Pts)**
2. Show by contradiction that if f is injective then $S \subset A \cup B$.**(02 Pts)**
3. Show by contradiction that if f is surjective then $A \cap B \subset \emptyset$.**(02 Pts)**
4. Show quickly that $A \cup B \subset S$ and $\emptyset \subset A \cap B$.**(01 Pts)**
5. Using the direct proof, show that if $A \cup B = S$ then f is injective.**(01 Pts)**
6. Using the direct proof, show that if $A \cap B = \emptyset$ then f is surjective.**(01 Pts)**
7. Using the previous questions, give the necessary condition to f to be a bijective function, then calculate f^{-1}.**(02 Pts)**